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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/016,890	08/07/2023	Yuta MARUO	18707-000053-US-NP	5901

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EXAMINER	
RAHLL, JERRY T	

ART UNIT	PAPER NUMBER
2874	

NOTIFICATION DATE	DELIVERY MODE
07/10/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

troymailroom@hdp.com

Office Action Summary	Application No. 18/016,890	Applicant(s) MARUO et al.	
	Examiner JERRY RAHLL	Art Unit 2874	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☐ Responsive to communication(s) filed on _____.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-7 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-7 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Information Disclosure Statement

2. The information disclosure statement (IDS) submitted on 19 January 2023 complies with the provisions of 37 CFR 1.97. Accordingly, the examiner has considered the information disclosure statement; please see attached form PTO-1449.

Drawings

3. The drawings submitted have been reviewed and determined to facilitate understanding of the invention. The drawings are accepted as submitted.

Claim Rejections - 35 USC § 103

4. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

5. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

6. This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly

owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

7. Claims 1-7 are rejected under 35 U.S.C. 103 as being unpatentable over Chinese publication CN 102411181 to Yan et al. ("CN1") in view of Japanese publication JP 2005107256 to Okada et al. ("JP1"), both cited in Applicant's IDS.

8. Regarding Claim 1, CN1 describes an optical fiber cable (see Figs 3-5) in which a plurality of drop optical cables ("butterfly shaped optical cables", see [0038] of submitted English language translation) are bundled. CN1 does not describe the particular structure of the optical cable. JP1 describes a butterfly-shaped optical cable having, in a cross section perpendicular to a longitudinal direction thereof, two or more neutral planes minimalizing a second moment of area (embodiment of Fig 5). Before the effective filing date of the claimed invention, it would have been obvious to one of ordinary skill in the art to use the cable of JP1 as the butterfly shaped optical drop cables of CN1. The motivation for doing so would have been to provide particular structure well-known in the art to the generally-described cable of CN1.

9. Regarding Claim 2, JP1 describes the drop cable comprising an optical fiber core (11) arranged at the center thereof, a sheath (12), covering a periphery of the optical fiber core, and a tension member (20) embedded in the sheath symmetrically with respect to the neutral planes (see Fig 5).

10. Regarding Claim 3, CN1 describes N^2 (where N = 1, 2, 3, ...) drop optical cables are arranged in a square shape on a cross section of the optical fiber cable, and each of the neutral planes of the plurality of drop optical cables is parallel to sides of the square shape (see Fig 3).

11. Regarding Claim 4, CN1 describes $2N^2-2N+1$ (where N = 1, 2, 3, ...) drop optical cables are arranged in a rhomboid shape on a cross section of the optical fiber cable, and each of the neutral planes of the plurality of drop optical cables is parallel to a diagonal line of the rhomboid shape (see Fig 3)

12. Regarding Claim 1, CN1 describes the N (where N > 3) drop optical cables are uniformly arranged in an outer periphery around tension member, and the neutral planes of the N drop optical cables are orthogonal to a straight line connecting the center of the drop optical cables and the center of the tension member (see Fig 4).

13. Regarding Claim 6, CN1 does not describe the N drop optical cables arranged in S-twist or SZ-twist around the tension member. However, S or SZ winding of sub-cables in an optical cable are well-known in the art for reducing stress on the sub-cables and their optical fibers. Before the effective filing date of the claimed invention, it would have been obvious to one of ordinary skill in the art to arrange the drop optical cables of the obvious optical fiber cable in view of CN1 and JP1. The motivation for doing so would have been to reduce stresses on the optical fibers.

14. Regarding Claim 7, JP1 describes drop cable have a cross-sectional shape regarded as a regular polygon, and a tension member is disposed at each corner (see Fig 5).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JERRY RAHLL whose telephone number is (571)272-2356. The examiner can normally be reached M-F 9:00am-5:00pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Uyen-Chau Le can be reached at 571-272-2397. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JERRY RAHLL/
Primary Examiner, Art Unit 2874